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5th Sem / Mechanical Engineering
Sub : Machine Design

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

Q.1 Shock resistance of steel is increased by adding.
(Co2)

- a) Nickel
- b) Chromium
- c) Nickel & Chromium
- d) Sulphur, Lead and Phosphorus

Q.2 The ratio of ultimate stress to design stress is called. (CO1)

- a) Bulk modulus
- b) Young's modulus of elasticity
- c) Factor of safety
- d) Poisson's ratio

Q.3 For designing ductile materials, which of the following theories used. (CO4)

- a) Maximum shear stress theory
- b) Shear strain energy theory
- c) Both 'a' & 'b'
- d) None of the above

- Q.4 While designing shafts, type of loading on the shafts may be (CO5)
- a) Twisting
 - b) Bending
 - c) Combination of twisting and bending moments
 - d) All of the above
- Q.5 One of the following which one is better method of making a shaft (CO5)
- a) Cold rolling
 - b) Hot rolling
 - c) Cold drawing
 - d) Machine turning
- Q.6 Which key transmits power through frictional resistance only (CO6)
- a) Woodruff key
 - b) Sunk key
 - c) Saddle key
 - d) Flat key
- Q.7 Riveted joints mostly fail by (CO6)
- a) Crushing of rivets
 - b) Bending of plates
 - c) Tearing of plates
 - d) Shearing of rivets
- Q.8 A Screw is specified by its (CO6)
- a) Major diameter
 - b) Minor diameter
 - c) Pitch diameter
 - d) Pitch
- Q.9 In kunckle joint pin is likely to fail in (CO6)
- a) Tension
 - b) Compression
 - c) Shear
 - d) Double Shear
- Q.10 The material of hub is flange coupling is (CO3)
- a) Cast Iron
 - b) Wrought Iron
 - c) High carbon steel
 - d) Low carbon steel

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 The ratio of ultimate stress to design stress is known as _____. (CO1)
- Q.12 The property of a material to be drawn into wires is called _____. (CO2)
- Q.13 Maximum normal stress theory is used for _____ materials. (CO4)
- Q.14 Define equivalent bending moment. (CO5)
- Q.15 The type of stresses developed in the keys are _____. (CO6)
- Q.16 What is permanent joint? (CO6)
- Q.17 Define Fullering. (CO5)
- Q.18 In square thread the flanks make an angle of _____ with axis. (CO6)
- Q.19 Write the application of power screws. (CO3)
- Q.20 Define flange coupling. (CO6)

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Define creep and fatigue. (CO2)
- Q.22 Define stress concentration? What are the factors used for reducing stress concentration. (CO3)
- Q.23 Explain is the maximum strain energy theory. (CO4)
- Q.24 Classify and explain different types of shafts. (CO5)
- Q.25 Sketch and differentiate sunk and saddle key. (CO6)
- Q.26 Differentiate between temporary and permanent joints. (CO6)
- Q.27 Differentiate between a knuckle joint and cutter joint. (CO6)

- Q.28 Drive the expressions for the strength of circular fillet weld subjected to bending moment. (CO6)
- Q.29 Explain the purpose and types of coupling. (CO5)
- Q.30 Give reasons of joining shaft with couplings. (CO5)
- Q.31 Discuss the design procedure of a spigot and socket cutter joint. (CO6)
- Q.32 Explain design procedure of muff coupling. (CO5)
- Q.33 Differentiate between lap joint and butt joint in riveted joints. (CO6)
- Q.34 Differentiate between parallel and transverse filled weld joint. (CO6)
- Q.35 Write the characteristics of good designer. (CO1)

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 A solid shaft is transmitting 1MW at 240 r.p.m.. Determine the diameter of the shaft if the maximum torque transmitted exceeds the mean torque by 20% Take the maximum allowed shear stress 60 N/mm^2 (CO5)
- Q.37 A double riveted lap joint is to be designed for 16mm thick plate. The riveting is of zig-zag type. The following permissible stresses may be assumed tensile stress = 110 MPa shear stress = 75 MPa and crushing stress = 150 MPa. State how the joint will fail and find the efficiency of joint. (CO6)
- Q.38 Write screw threads nomenclature with diagram. (CO6)